

Title: APT Profiling and Detection in Critical Infrastructure using Deception Tech

Group No : 10

Guide : Mr. Sreejith K

ABSTRACT

Critical infrastructure is a prime target for Advanced Persistent Threats (APTs), which are notoriously difficult to detect as they mimic legitimate network activity. Traditional security tools often fail to identify these stealthy attacks, resulting in significant detection delays and increased security risks. This project's primary objective is to bridge this critical detection gap by developing a robust framework for rapidly identifying and analyzing APT behavior within these essential networks.

Our proposed solution employs a proactive defense model centered on deception technology. By deploying a strategically designed network of decoys, traps, and fake digital assets, our framework lures adversaries into a controlled and monitored environment. Any interaction with these deceptive elements generates clear, high-fidelity alerts, immediately exposing the attackers' presence and actions.

The potential impact of this approach is a significant improvement in threat detection speed and accuracy. It enables security teams to reduce false positives, gain a deeper understanding of attacker behavior, and ultimately strengthen the cyber resilience and defense posture of critical infrastructure against sophisticated, advanced cyberattacks.